

Applied Statistical Programming

Week 3

Why do we need the bootstrap?

Suppose we observe a sample

$$X = \{X_1, \dots, X_n\} \sim F_0$$

and compute a statistic of interest

$$T(X).$$

For inference, we need the sampling distribution of $T(X)$:

$$G_n(t, F_0) = \Pr\{T(X) \leq t\}.$$

- This distribution could be a complicated function of the unknown population distribution F_0 .
- Without it, we cannot get standard measures to assess uncertainty.

Two routes

The inferential problem is always the same:

$G_n(t, F_0)$ depends on an unknown F_0 .

Asymptotics

- Let $n \rightarrow \infty$.
- Use results such as the CLT and delta method.
- Replace $G_n(t, F_0)$ with a limiting distribution that no longer depends on F_0 .

Bootstrap

- Keep the finite-sample statistic.
- Replace the unknown F_0 with the empirical distribution $\hat{F}_n$.
- Use simulation to approximate $G_n(t, \hat{F}_n)$.

The empirical distribution

We define the empirical distribution of F_0 as:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq x).$$

If the target parameter is a functional $\theta = T(F)$, the plug-in estimator is

$$\hat{\theta} = T(\hat{F}_n).$$

The bootstrap idea

The bootstrap approximation is

$$G_n(t, F_0) \approx G_n(t, \hat{F}_n),$$

The bootstrap asks: what would sampling variability look like if $\hat{F}_n$ were the population?

The logic requires:

- 1 $\hat{F}_n$ is close to F_0 when n is large.
- 2 The distribution of the statistic changes smoothly as F changes.

Sampling with replacement from the data is just simulation from $\hat{F}_n$. As a result:

- By virtue of design: an analytic sampling-distribution problem becomes a computational one.

Nonparametric bootstrap

The goal of the bootstrap is to approximate the sampling distribution

$$\mathcal{L}_F(\hat{\theta} - \theta)$$

using only the observed sample.

For $b = 1, \dots, B$:

❶ Sample

$$X_1^{*(b)}, \dots, X_n^{*(b)} \stackrel{iid}{\sim} \hat{F}_n,$$

equivalently, sample with replacement from $\{X_1, \dots, X_n\}$.

❷ Compute

$$\hat{\theta}^{*(b)} = T(\hat{F}_n^{*(b)}).$$

Where does bootstrap approximation error come from?

The bootstrap replaces an unknown object with an observable one:

$$\mathcal{L}_F(\hat{\theta} - \theta) \longrightarrow \mathcal{L}_{\hat{F}_n}(\hat{\theta}^* - \hat{\theta}).$$

Two distinct approximations.

- 1 **Bootstrap approximation:** we replace the unknown population distribution F with the empirical distribution $\hat{F}_n$.

$$\mathcal{L}_F(\hat{\theta} - \theta) \approx \mathcal{L}_{\hat{F}_n}(\hat{\theta}^* - \hat{\theta}).$$

This approximation depends on n and on whether the bootstrap is valid for the statistic T .

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- 2 **Monte Carlo approximation:** we cannot usually calculate the bootstrap distribution exactly, so we approximate it using only B bootstrap replications.

$$\mathcal{L}_{\hat{F}_n}(\hat{\theta}^* - \hat{\theta}) \approx \left\{ \hat{\theta}^{*(1)} - \hat{\theta}, \dots, \hat{\theta}^{*(B)} - \hat{\theta} \right\}.$$

Increasing B reduces only this second source of approximation error.

Example: sampling error of the median

Let

$$M_n = \text{median}(X_1, \dots, X_n)$$

be the sample median, an estimator of the population median.

The same questions we ask about the mean also apply here:

- What is the variance of M_n ?
- How can we construct a confidence interval for the population median?

Unlike the sample mean, the median does not come with a simple plug-in standard error formula in introductory settings.

Example: medians

Start with observed data $X_1, \dots, X_n$ and compute

$$M_n = \text{median}(X_1, \dots, X_n).$$

For each bootstrap replication $b = 1, \dots, B$:

- 1 Draw $X_1^{*(b)}, \dots, X_n^{*(b)}$ with replacement from $\{X_1, \dots, X_n\}$.
- 2 Compute

$$M_n^{*(b)} = \text{median}(X_1^{*(b)}, \dots, X_n^{*(b)}).$$

The collection $M_n^{*(1)}, \dots, M_n^{*(B)}$ approximates the sampling behavior of M_n .

Example: medians

The bootstrap variance estimate is the sample variance of the bootstrap medians:

$$\widehat{\text{Var}}_B(M_n) = \frac{1}{B-1} \sum_{b=1}^B \left(M_n^{*(b)} - \overline{M^*} \right)^2,$$

where

$$\overline{M^*} = \frac{1}{B} \sum_{b=1}^B M_n^{*(b)}.$$

A normal-approximation bootstrap interval uses the bootstrap standard error:

$$M_n \pm q_{1-\alpha/2} \sqrt{\widehat{\text{Var}}_B(M_n)},$$

Here $q_{1-\alpha/2}$ is the standard normal quantile. The interval mirrors the familiar confidence interval for a mean, but replaces the analytic standard error with a bootstrap estimate.

Beyond the median

The empirical bootstrap is not tied to one statistic.

Once the statistic can be recomputed on a resampled dataset, the same workflow applies:

- 1 Resample observations from the empirical distribution.
- 2 Recompute the statistic.
- 3 Use the bootstrap distribution to estimate uncertainty.

Examples include quantiles, interquartile ranges, skewness, kurtosis, regression coefficients, and many other statistics.

Bootstrap bias estimate

The bootstrap estimate of finite-sample bias is

$$\widehat{\text{bias}}_{\text{boot}} = \mathbb{E}^*(\hat{\theta}^*) - \hat{\theta} \approx \overline{\theta^*} - \hat{\theta}.$$

A simple bias-corrected point estimator is

$$\hat{\theta}_{\text{bc}} = \hat{\theta} - \widehat{\text{bias}}_{\text{boot}} = 2\hat{\theta} - \overline{\theta^*}.$$

Let q_{α}^* denote the α quantile of bootstrap replicates.

Percentile interval

$$[q_{\alpha/2}^*, q_{1-\alpha/2}^*].$$

Biased-corrected interval

$$[2\hat{\theta} - q_{1-\alpha/2}^*, 2\hat{\theta} - q_{\alpha/2}^*].$$

When is the ordinary bootstrap plausible?

At a high level, we want the empirical distribution to approximate the relevant local behavior of F well enough that

$$\sup_t \left| \Pr \left(\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \leq t \right) - \Pr \left(\sqrt{n}(\hat{\theta} - \theta) \leq t \right) \right| \xrightarrow{p} 0.$$

- Smooth statistics of i.i.d. data often behave well.
- Nonsmooth boundaries, extreme order statistics, weak identification, or dependence may require modified bootstrap schemes.
- Computation cannot repair an invalid resampling design.

Bootstrap failure: Anderson (1997, *Econometrica*)

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} F, \quad E[X_i] = \mu, \quad \text{Var}(X_i) = 1,$$

but we know that

$$\mu \geq 0.$$

A natural estimator for the mean imposes this restriction, i.e.,

$$\hat{\mu}_n = [\bar{X}_n]_+ := \max\{0, \bar{X}_n\}.$$

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Suppose that $\mu = 0$, this means that:

$$\sqrt{n}(\hat{\mu}_n - \mu) = \sqrt{n}([\bar{X}_n]_+ - \mu) \xrightarrow{d} [G]_+, \quad G \sim N(0, 1).$$

The last line is a direct result of applying the CLT and the preservation under continuous mappings theorem.

Why does the ordinary bootstrap fail?

The bootstrap counterpart of $\sqrt{n}(\hat{\mu}_n - \mu)$ is:

$$\sqrt{n}(\hat{\mu}_n^* - \hat{\mu}) = \sqrt{n}([\bar{X}_n^*]_+ - \bar{X}_n)_+$$

This quantity does not converge to $[G]_+$. To see this, consider a constant $c > 0$, then we have:

$$\begin{aligned}\Pr(\sqrt{n}(\hat{\mu}_n^* - \hat{\mu}) \leq x | \sqrt{n}(\bar{X}_n) > c) &= \Pr(\sqrt{n}([\bar{X}_n^*]_+ - \bar{X}_n)_+ \leq x | \sqrt{n}(\bar{X}_n) > c) \\ &= \Pr(\sqrt{n}(\max\{0, \bar{X}_n^*\} - \max\{0, \bar{X}_n\}) \leq x | \sqrt{n}(\bar{X}_n) > c) \\ &= \Pr(\max\{\sqrt{n}(\bar{X}_n^* - \bar{X}_n), -\sqrt{n}\bar{X}_n\} \leq x | \sqrt{n}(\bar{X}_n) > c) \\ &\geq \Pr(\max\{\sqrt{n}(\bar{X}_n^* - \bar{X}_n), -c\} \leq x | \sqrt{n}(\bar{X}_n) > c) \\ &\xrightarrow{P} \Pr(\max\{G, -c\} \leq x) \\ &\geq \Pr([G]_+ \leq x)\end{aligned}$$

Note that

- With nonvanishing probability, the bootstrap is operating from a sample in exactly this situation.
- For $-c < x < 0$ the last inequality is strict.

Clustered data: resample at the dependence level

If observations are dependent within clusters $g = 1, \dots, G$, individual-row resampling destroys that dependence.

A cluster bootstrap samples clusters:

$$g_1^*, \dots, g_G^* \stackrel{iid}{\sim} \{1, \dots, G\},$$

then includes all observations belonging to sampled clusters.

- The computational unit mirrors the stochastic independence unit.
- This same principle guides parallel decomposition: parallelize over independent replicates, not dependent substeps.

Permutation tests

- Parametric tests (e.g., the t -test and the F -test) get their null distribution from distributional assumptions (normality, known asymptotic form, etc.)
- Permutation tests instead build the null distribution (H_0 directly from the data:
 - ▶ Require *exchangeability* under H_0
 - ▶ Work for essentially any statistic: differences in means, medians, ranks, regression coefficients
 - ▶ Are exact (finite-sample valid, not just asymptotically) when exchangeability holds

What we permute?

- The null hypothesis determines what we permute
- For each unit $i = 1, \dots, n$, let:
 - ▶ Y_i denote the observed outcome
 - ▶ $D_i \in \{0, 1\}$ the group label (e.g., treatment vs. control)
 - ▶ $T(Y, D)$ be any statistic computed from them
- Under H_0 , labels are **exchangeable**: for any permutation $\pi \in \Pi$,

$$T(Y, \pi D) \stackrel{d}{=} T(Y, D).$$

- ▶ In randomization inference: exchangeability is a product of the experiment
- ▶ In observational inference: permutation tests require stronger assumptions

Example 1

- Take $n = 4$ units with outcomes $Y = (Y_1, Y_2, Y_3, Y_4)$ and the observed treatment labels $D = (1, 1, 0, 0)$
- A permutation π just relabels which units are called “treated,” holding Y fixed:

	Unit 1	Unit 2	Unit 3	Unit 4
Observed D	1	1	0	0
$\pi_1 D$	1	0	1	0
$\pi_2 D$	0	1	0	1
$\pi_3 D$	0	0	1	1

- ▶ With $n = 4$ and two units per group, there are $\binom{4}{2} = 6$ distinct arrangements of D in Π
- ▶ For each $\pi \in \Pi$, recompute $T(Y, \pi D)$ using the *same* outcomes Y but the relabeled πD
- ▶ The observed assignment D itself is one element of Π

Example 2

- Suppose we observe two groups:

$$Y_1^{(1)}, \dots, Y_m^{(1)} \quad \text{and} \quad Y_1^{(0)}, \dots, Y_n^{(0)}.$$

- To test whether the groups differ:
 - 1 Pool all $m + n$ observations.
 - 2 Randomly assign m observations to Group 1.
 - 3 Assign the remaining n observations to Group 0.
 - 4 Recompute the statistic, for example

$$T^* = \bar{Y}^{*(1)} - \bar{Y}^{*(0)}.$$

- 5 Repeat over $\pi \in \Pi$ (exactly or by simulation) to build the null distribution of T^* .

Exact permutation p -value

- Let Π be the set of permutations. For a two-tailed test,

$$p_{\text{exact}} = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbb{I}\{|T(Y, \pi D)| \geq |T(Y, D)|\}.$$

- If all $\binom{n}{n_1}$ treatment assignments are equally likely under the design, enumerate them when possible.
 - ▶ The reference distribution comes from the null assignment mechanism, not an asymptotic approximation.
 - ▶ Enumeration is exact but grows combinatorially in n

Monte Carlo permutation p -value

- When $|\Pi|$ is incredibly large, sample B random permutations. Then calculate:

$$\hat{p} = \frac{\sum_{b=1}^B \mathbb{I}\{|T_b^*| \geq |T_{\text{obs}}|\}}{B}.$$

- ▶ Conditional on the observed data and test statistic, approximation error decreases at rate $B^{-1/2}$.
- ▶ In practice: this is what you use whenever exact enumeration is not possible.

To wrap up:

Permutation and bootstrap procedures resample for different purposes.

	Permutation test	Bootstrap
Sampling	Without replacement	With replacement
Center	Null hypothesis	Observed estimate
Goal	p -value	Standard error / CI

- Use a **permutation test** when you have a sharp null to test.
- Use the **bootstrap** when you need a standard error or confidence interval around an estimate, with no sharp null in play.